

Problem-Solution Fit Canvas

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Problem–Solution Fit Canvas helps validate whether the proposed Emotion Detection and Learning Support Engine effectively addresses the real needs of students. It identifies the target users, their challenges, emotional experiences, existing alternatives, and how the proposed AI-powered solution creates value through personalized emotion detection and learning guidance.

1. Customer Segments [CS] College students, School students, Online learners, Self-paced learners, Learners preparing for competitive examinations	2. Problems / Pains [PR] Difficulty concentrating, Stress/ anxiety, Lack of personalized support. Frequency: High	3. Triggers to Act [TR] Poor performance, Frustration/ confusion, Need for motivation, Exam preparation	4. Emotions [EM] Before: Confused, Stressed, Frustrated, Anxious After: Motivated, Confident, Calm, Focused
5. Available Solutions [AS] Generic e-learning, Online portals, Counseling. Pros: Easy access. Cons: No emotion-aware support, Generic.	6. Customer Limitations [CL] Limited time, Budget, Laptop/ smartphone only, Internet dependency	7. Behavior & Intensity [BE] Behavior: Search online, Practice, Discouraged after failure. Intensity: Med-High	8. Channels of Behavior [CH] Online: Streamlit app, Educational sites Offline: Classroom, Books, Teacher discussions
9. Problem Root Cause [RC] Platforms ignore emotional state; Lack of personalized guidance; Students hesitate to seek help.			
10. Your Solution [SL] AI-powered Engine: Detects emotions via BiLSTM, Personalized guidance, Confidence scoring, Prediction history storage.			